

Analytix.pk-Data Analyst with AI + AWS + Snowflake (Cohort-17)

Assignment-01

Detailed Business Insights

A. Product-Centric Insights:

1. Which products generate the highest revenue vs highest quantity?

The **highest revenue product** is **Dotcom Postage** with about **\$206,245**, but it has only around **707 units**. This means it is not a high-volume product; it contributes revenue because of high unit value/service charge.

The **highest quantity product** is **World War 2 Gliders Asstd. Designs** with about **53,847 units**, but its revenue is only around **\$13,588**. So it sells in very high volume, but at a low unit value.

2. Are top-selling products also high-revenue contributors?

Not always. Some products appear in both high quantity and high revenue, such as:

- **Jumbo Bag Red Retrospot**
- **White Hanging Heart T-Light Holder**
- **Assorted Colour Bird Ornament**
- **Rabbit Night Light**

These are strong business-driving products because they combine both **high demand** and **high revenue contribution**.

But some high-quantity products, such as **World War 2 Gliders Asstd Designs**, generate much lower revenue, meaning they are likely low-price products.

3. Which products are frequently bought together?

Based on invoice-level product combinations, some commonly bought-together products are:

- **Jumbo Bag Pink Polkadot + Jumbo Bag Red Retrospot**
- **Jumbo Bag Red Retrospot + Lunch Bag Red Retrospot**
- **Party Bunting + Spotty Bunting**
- **60 Teatime Fairy Cake Cases + Pack of 72 Retrospot Cake Cases**
- **Party Bunting + Regency Cakestand 3 Tier**

This suggests customers often buy products from similar themes or usage categories together, such as bags, party items, cake cases, and decoration products.

4. Are there “low quantity but high revenue” premium products?

Some products have relatively low quantity but high revenue, for example:

- **Picnic Basket Wicker 60 Pieces** - about **61 units**, but around **\$39,620 revenue**
- **Rotating Silver Angels T-Light Hldr** - about **100 units**, but around **\$26,142 revenue**
- **Vintage Red Kitchen Cabinet** - about **50 units**, but around **\$6,395 revenue**
- **Rustic Seventeen Drawer Sideboard** - about **32 units**, but around **\$5,000 revenue**

These look like premium or high-value products.

Final Product Insight

The products that actually drive business growth are not only the highest-quantity products. The strongest products are those that combine **high sales volume and strong revenue**, especially **Jumbo Bag Red Retrospot, White Hanging Heart T-Light Holder, Rabbit Night Light, and Assorted Colour Bird Ornament**. Premium products and postage-related revenue also contribute significantly, but they should be interpreted separately from normal product sales.

B. Product-Centric Insights:

1. Do we have repeat transactions or bulk purchasing patterns?

Yes. The dataset shows clear repeat and bulk purchasing behavior.

There are around **25,900 unique invoices**. Out of **4,372 customers**, around **3,059 customers** made more than one transaction, which means about **70% of identified customers are repeat buyers**.

There is also strong evidence of bulk purchasing. Around **50% of invoices contain 100 or more total units**, and about **51% of invoices contain 10 or more product lines**. This suggests many customers are not buying single items only; they are purchasing baskets or bulk orders.

2. What is the average revenue per invoice?

- Total revenue is approximately: **\$9,747,747.93**
- Total unique invoices: **25,900**
- So average revenue per invoice is approximately: **\$376.36 per invoice**
- If cancelled/negative invoices are excluded, the average positive invoice value increases to around: **\$534.40 per invoice**

3. Are there spikes in transaction volume on specific days/months?

Yes. The strongest monthly spike is in **November 2011**.

Highest invoice-volume months:

Month	No. of Invoices	Revenue (\$)
Nov-2011	3,462	1,461,756.25
Oct-2011	2,637	1,070,704.67
Sep-2011	2,327	1,019,687.62

This shows a strong increase in sales activity from **September to November**, likely due to seasonal or holiday-related purchasing.

4. Do weekends vs weekdays impact transaction size?

Yes. Weekdays perform stronger than weekends.

There are no Saturday transactions in the dataset. Sunday has fewer transactions and lower average invoice value.

Day Type	Invoices	Revenue (\$)	Avg. Revenue (\$)/Invoice
Weekdays	23,519	8,942,069.04	380.21
Sunday	2,381	805,678.89	338.38

So, weekday transactions are more frequent and slightly higher in value.

Final Transaction Insight

Customers show strong repeat and bulk-buying behavior. The business is heavily driven by larger baskets and repeat transactions, especially during the September-November period. November 2011 is the strongest transaction month, while weekdays, especially **Thursday and Tuesday**, generate the highest sales activity.

C. Customer-Centric Insights:

1. Who are the top revenue-generating customers?

The top customer is the one which generated about **\$1,447,682.12** in total revenue although it's Customer ID is not identified. The second highest customer is with **ID 14646**, generating about **\$279,489** in total revenue.

Top customers include:

Customer ID	Revenue (\$)
Missing Customer ID	1,447,682.12
14646	279,489.02
18102	256,438.49
17450	187,482.17
14911	132,572.62
12415	123,725.45

These customers are very important because they contribute a large share of total sales.

2. Do a small group of customers contribute most revenue?

Yes. The customer base follows a **Pareto-style pattern**.

There are **4,372 identified customers**. The top **20% of customers** contribute around **74% of identified customer revenue**. Also, only about **1,171 customers**, around **27% of customers**, are needed to generate nearly **80% of revenue**.

This means revenue is highly concentrated among a smaller group of valuable customers.

3. Are customers one-time buyers or repeat buyers?

Out of **4,372 customers**:

Customer Type	Number of Customers
One-time buyers	1,313
Repeat buyers	3,059

So around **70% of customers are repeat buyers**.

This is a positive sign because the business is not only dependent on one-time purchases. Many customers return and purchase again.

4. Which customers purchase high-value baskets?

Some customers place very large basket orders. Examples include:

Customer ID	Highest Basket Revenue (\$)
16446	168,469.60
12346	77,183.60
15098	38,970.00
17450	31,698.16
12415	22,775.93

Customer **16446** is especially interesting because they placed only a few transactions but generated a very high basket value.

Final Customer Insight

The revenue engine is mainly driven by **repeat customers and a small group of high-value buyers**. Customers such as **14646, 18102, 17450, 14911, and 12415** should be treated as key accounts because they contribute strongly to total revenue. The business should focus on retaining these customers while also trying to convert one-time buyers into repeat buyers.

D. Country / Region-Centric Insights:

1. Which countries contribute the highest revenue?

The business is strongest in the **United Kingdom**, which generates about **\$8.19 million**, equal to around **84% of total revenue**.

Top revenue countries:

Country	Revenue (\$)	Quantity
United Kingdom	8,187,806	4,263,829
Netherlands	284,662	200,128
EIRE	263,277	142,637
Germany	221,698	117,448

Country	Revenue (\$)	Quantity
France	197,404	110,480

2. Is revenue concentrated or evenly distributed?

Revenue is **highly concentrated**.

The **United Kingdom alone contributes around 84%** of total revenue. The top 5 countries together contribute around **94%** of total revenue.

This means the business is not geographically balanced. It depends heavily on the UK market.

3. Which countries have high quantity but low revenue?

Some countries sell high quantities but generate lower revenue per unit.

Examples:

Country	Quantity	Revenue (\$)	Revenue per Unit (\$)
Sweden	35,637	36,596	1.03
Japan	25,218	35,341	1.40
Netherlands	200,128	284,662	1.42
Australia	83,653	137,077	1.64

These countries may be buying lower-priced products or bulk discounted items.

4. Which regions show growth potential?

The strongest growth-potential countries are:

- **Netherlands** - high quantity and second-highest revenue after UK.
- **EIRE** - strong revenue and good order volume.
- **Germany and France** - both have high transaction volume and stable revenue contribution.
- **Australia** - strong quantity and decent revenue, but lower revenue per unit.

These markets are already active and could be expanded through targeted product promotion, customer retention, or premium product offers.

Final Country Insight

The business is strongest in the **United Kingdom**, but this also creates risk because revenue is highly dependent on one country. Outside the UK, the most important markets are **Netherlands, EIRE, Germany, France, and Australia**. Countries such as **Sweden, Japan, and Netherlands** show high quantity but lower revenue per unit, suggesting opportunities to introduce higher-value products or reduce discount dependency.

E. Time-Based Insights (Trend Analysis):

1. Is monthly revenue growing, declining, or seasonal?

Monthly revenue shows a **seasonal pattern**. Revenue is moderate during the first half of 2011, then increases strongly from **September to November 2011**.

This suggests higher sales activity before the year-end/holiday period.

2. Which months show peak performance?

The strongest month is **November 2011**, with approximately **\$1.46 million** in revenue.

Top months:

Month	Revenue (\$)	Invoices
Nov-2011	1,461,756	3,462
Oct-2011	1,070,705	2,637
Sep-2011	1,019,688	2,327
Dec-2010	748,957	2,025
May-2011	723,334	2,162

3. Which weekdays show highest sales activity?

Thursday is the strongest weekday, generating about **\$2.11 million** in revenue and **5,660 invoices**.

Weekday	Revenue (\$)	Invoices
Thursday	2,112,519	5,660
Tuesday	1,966,183	4,722

Weekday	Revenue (\$)	Invoices
Wednesday	1,734,147	4,815
Monday	1,588,609	4,138
Friday	1,540,611	4,184
Sunday	805,679	2,381

There are no Saturday transactions in the dataset.

4. Are there sudden spikes or drops?

Yes. Major revenue spikes:

- **May 2011** increased by about **47%** compared with April.
- **September 2011** increased by about **49%** compared with August.
- **November 2011** increased by about **37%** compared with October.

Major drops:

- **April 2011** dropped by about **28%** compared with March.
- **December 2011** dropped sharply, but this may be because the dataset contains only part of December, not the full month.

Final Time-Based Insight

The business performs best during the **September–November period**, with **November 2011** being the peak month. On a weekday basis, **Thursday and Tuesday** are the strongest sales days. The trend suggests strong seasonal demand before year-end, while Sunday has lower activity and Saturday has no recorded transactions.